

$K_{3,3}$ IS A COCOMPARABILITY GRAPH WITH NO P_a -AVOIDING VERTEX ORDER

ABSTRACT. Feuilloley and Habib classify grounded intersection graph classes by forbidden ordered patterns, and ask (their Open problem 2) whether the class C_a — the graphs admitting a vertex order with no occurrence of the four-vertex ordered pattern P_a — and the cocomparability graphs are incomparable, offering as the alternative that the cocomparability graphs are contained in C_a . We settle the inclusion the question leaves open: $K_{3,3}$ is a permutation graph, hence a cocomparability graph, and none of its $6! = 720$ vertex orders avoids P_a , so the cocomparability graphs are not contained in C_a and the alternative offered in the problem’s parenthetical is false. The argument is short and uses no computer. The other direction — that the reverse inclusion fails, witnessed by C_6 — is asserted by the source itself and is taken from there; so incomparability follows, but only one of its two halves is proved here. Since the cocomparability graphs are interval filament graphs, the inclusion $C_a \subseteq$ interval filament graphs, singled out in the paper as the lowest edge of its diagram, is strict.

1. THE QUESTION

Feuilloley and Habib [2] classify classes of grounded intersection graphs by forbidden *ordered* patterns. For $S \subseteq \{a, b, c, d\}$ they define, verbatim:

“Let $S \subseteq \{a, b, c, d\}$, the pattern P_S is the pattern on four vertices, with edges $(1, 3)$ and $(2, 4)$, and non-edge set S . The class C_S is the class where P_S is forbidden.”

The four optional pairs are named $a = (1, 2)$, $b = (2, 3)$, $c = (3, 4)$, $d = (1, 4)$ on the immediately preceding line of the same source, and by its drawing convention (“plain edges for edges, dashed edges for non-edges, and nothing for undecided edges”) a pair outside S is unconstrained. Concretely:

P_a is the ordered pattern on $v_1 < v_2 < v_3 < v_4$ with $v_1v_3 \in E$, $v_2v_4 \in E$ and $v_1v_2 \notin E$, the three pairs $(2, 3)$, $(3, 4)$, $(1, 4)$ being unconstrained; and $G \in C_a$ iff some linear order of $V(G)$ contains no four vertices realizing P_a .

The same definition, with the same names for the four pairs, is repeated by four of the same authors in [1], so the reading above is not an inference from a single sentence.

The paper’s second open problem is stated verbatim:

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“Are C_a and cocomparability graphs incomparable? (Or are cocomparability graphs included in C_a , the reverse being impossible, as witnessed by C_6 ?)”

and its lead-in says why it is asked (the missing word after “if” is the source’s own, quoted unrepaired):

“Of course settling the status of each inclusion (either equality or strict inclusion) is a natural direction. For example, the lowest such edge in the diagram is for the inclusion of C_a in interval filament graphs. This question would be solved if could answer positively the following, innocent-looking problem.”

The statement unfolds into two existentials. That some member of C_a is not a cocomparability graph is the source’s own, witnessed by C_6 : the cycle order $1, 2, \dots, 6$ has no crossing pair of edges at all, hence no P_a , while C_6 is not a cocomparability graph by Gallai [5], the citation the source itself uses. The open content is the other direction, and it is what Theorem 1 settles.

Remark. Two conventions of the source are recorded because they could otherwise be mistaken for mathematics. First, the paper writes the class both as C_a and as C_a ; the two denote the same class and we write C_a . Second, the source defines $\mathcal{C}_{\mathcal{F}}$ as “the set of **connected** graphs that have an ordering avoiding the patterns in $\mathcal{F}$ ”. Read literally that clause makes the open problem degenerate (the disconnected cocomparability graph $2K_2$ would settle it for a reason having nothing to do with patterns) and it also falsifies the paper’s own assertion that interval graphs are contained in C_a ; it is a normalisation. Nothing below turns on the choice, because the witness we exhibit is connected.

The same shape of statement is already published for a sibling pattern. Catanzaro, Chaplick, Felsner, B. V. Halldórsson, M. M. Halldórsson, Hixon and Stacho [4] characterise the max point-tolerance graphs by a vertex order (their Definition 5.3 and Theorem 5.4) requiring that for all $u < v < w < x$ with $uw, vx \in E$ one has $vw \in E$; that is avoidance of the four-vertex pattern with edges $(1, 3)$, $(2, 4)$ and non-edge $(2, 3)$, which is the class C_b of [2] (there identified with the grounded rectangle graphs, of which max point-tolerance is a synonym). Their Observation 6.9 states that this class “is incomparable with both planar graphs and permutation graphs”, the witness being the octahedron $K_{2,2,2}$, a permutation graph that is not a max point-tolerance graph; Jelínek and Töpfer [3] restate it as “neither Per nor any superclass of Per is contained in Mpt”. Neither work addresses C_a , and neither settles Open problem 2: their statement is about C_b , and their witness does not transfer, since [4] also states that every complete bipartite graph is a 2D ray graph and that 2D ray graphs are max point-tolerance graphs, which on their account puts $K_{3,3}$ inside C_b ; we do not re-derive either statement.

2. THE WITNESS

Let $G = K_{3,3}$ with parts $A = \{0, 1, 2\}$ and $B = \{3, 4, 5\}$, so $n = 6$, $m = 9$ and

$$E(K_{3,3}) = \{03, 04, 05, 13, 14, 15, 23, 24, 25\},$$

$$\overline{K_{3,3}} = \{01, 02, 12\} \cup \{34, 35, 45\} = 2K_3.$$

Theorem 1. *The cocomparability graphs are not contained in C_a , and the failure already occurs inside the permutation graphs: $K_{3,3}$ is a permutation graph, hence a cocomparability graph, and none of the 720 linear orders of $V(K_{3,3})$ avoids P_a .*

Proof that $K_{3,3}$ is a cocomparability graph. The inversions of $\pi = 456123$ are exactly the nine pairs $\{i, j\}$ with $i \in \{1, 2, 3\}$ and $j \in \{4, 5, 6\}$, so the inversion graph of π is $K_{3,3}$, which is therefore a permutation graph. By Pnueli, Lempel and Even [6] the permutation graphs are exactly the graphs that are simultaneously comparability and cocomparability graphs. $\square$

Proof that no order of $K_{3,3}$ avoids P_a . In $K_{3,3}$ two vertices are non-adjacent iff they lie in the same part. So an occurrence of P_a at positions $i < j < k < l$ needs v_i, v_j in one part, say A ; the edge $v_i v_k$ then forces $v_k \in B$ and the edge $v_j v_l$ forces $v_l \in B$. Hence

$$(1) \quad \begin{array}{c} \text{some two vertices of one part sit at positions} \\ P_a \text{ occurs} \iff i < j \text{ and two vertices of the other part both} \\ \text{sit at positions} > j. \end{array}$$

Fix an order and write $p_1 < p_2 < p_3$ for the positions of A and $q_1 < q_2 < q_3$ for those of B . Applying (1) to the pair (p_1, p_2) , avoidance allows at most one vertex of B after p_2 , so at least two vertices of B precede p_2 , i.e. $q_2 < p_2$. Applying it to (q_1, q_2) gives $p_2 < q_2$. These contradict each other, so every order contains P_a . $\square$

For a concrete occurrence, take the alternating order 031425 and the positions $1 < 3 < 4 < 6$, carrying the vertices 0, 1, 4, 5: the pair 01 is a non-edge and 04, 15 are edges.

Corollary 2. *C_a and the cocomparability graphs are incomparable, which is the answer asked for by Open problem 2 of [2].*

Proof. Theorem 1 gives one direction. The other is the source's own assertion, quoted in Section 1 and reproduced there: $C_6 \in C_a$ via the cycle order, and C_6 is not a cocomparability graph [5]. Only the direction of Theorem 1 is new here. $\square$

3. THE CONSEQUENCE FOR THE DIAGRAM OF [2]

The lead-in quoted in Section 1 singles out the inclusion of C_a in the interval filament graphs as the lowest edge of the paper's Figure 1, and observes that a positive answer to the open problem settles it.

Corollary 3. $C_a \subsetneq$ *interval filament graphs*.

Proof. The inclusion cocomparability $\subseteq$ interval filament is one of the items of the observation in [2] that collects the inclusions of its Figure 1; we do not re-derive it. Strictness is Theorem 1: $K_{3,3}$ is a cocomparability graph, hence an interval filament graph, and $K_{3,3} \notin C_a$. $\square$

4. THE COMPANION PROGRAM

The file `verify.py` accompanying this note checks the objects printed above: the printed edge list of $K_{3,3}$ against the printed parts, “0 of 720 orders of $K_{3,3}$ avoid P_a ” by two independent deciders, the pigeonhole step of the second proof on each of the 720 orders, the occurrence printed for the alternating order, and that the inversion graph of π is $K_{3,3}$; it also includes a check that would fail if the optional pairs a and b of the pattern were read the wrong way round, and checks of the two facts about C_6 that the source’s parenthetical asserts. It does not re-derive the inclusion cocomparability $\subseteq$ interval filament, which is quoted from [2].

The program is not confined to this note, and the reader should be warned. It is Python 3.9+, standard library only, and reports 43 checks, all passing, but fewer than half of them bear on the statements above; the rest verify results this note does not state — a membership criterion for complete multipartite graphs, that every complete multipartite graph is a permutation graph, that no graph on at most five vertices lies outside C_a , and further controls — and its section headings and scope note accordingly refer to numbered theorems and corollaries that do not appear here. Nothing above depends on any of those checks.

5. WHAT IS NOT SETTLED

No characterization of $C_a \cap$ cocomparability is offered, and no structural (non-pattern) description of C_a ; the latter is Open problem 3 of [2] and is untouched here. Nothing here bears on whether C_a has a finite forbidden induced subgraph characterization, and no such claim is made. $K_{3,3}$ is not claimed to be the only witness of its order. Corollary 2 is half quoted: the direction witnessed by C_6 is the source’s assertion and is not proved here. Corollary 3 inherits the inclusion cocomparability $\subseteq$ interval filament from [2] and does not re-derive it. Finally, the result is easy, and that is the honest framing: the decisive step is a short pigeonhole argument about a six-vertex graph, and its interest is the diagram consequence that [2] itself said the problem would buy.

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